

WORKPACK 1

Find Your Way Around

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do. Each tick shows how much of a head start you have.

If you can tick all eight: skip to the Mini-Mission on page 8 and prove it in two minutes. Then move on to Pack 2.

- ☐ I can open an Excel file from a folder or an email attachment.
- ☐ I can find the sheet tabs at the bottom of the Excel window and click between them.
- ☐ I know what the Name Box is, and I can type a cell address into it to jump there.
- ☐ I can press Ctrl+Home to jump straight to cell A1.
- ☐ I can press Ctrl+End to find the last row of data in a sheet.
- ☐ I can widen a column when the text inside it is cut off.
- ☐ I can find a sheet called Data Dictionary and use it to understand a column name.
- ☐ I can find out how many data records are in a file without scrolling through every row.

Marcus from the next desk says: "I just scroll down and count." There is a faster way. This pack shows you three of them.

What This Pack Is About

By the end of this pack, you will be able to open the GreenWheels hire dataset, move around it using keyboard shortcuts, and use the Data Dictionary sheet to understand what each column means. These are the navigation skills you need before you do any analysis.

The pack takes around 25 to 35 minutes if you work through every section. If you ticked most of the warm-up questions, it will take less. You can stop after any skill chunk and come back later — nothing needs to be done in one sitting.

The GreenWheels Dataset

The file you will use is `urban_cycle_hire_aggregated.xlsx`. It contains quarterly hire records from 10 cycle stations across the city. Here is a five-row sample so you can see what the data looks like before you open the full file.

Record ID	Station Name	Zone	Period	Start Date	Total Hires	Avg Duration (Mins)	Revenue (£)	Active Bikes	Satisfaction
HC-001	Central Square	Central	Q1 2024	01/01/2024	512	28	£4,352	25	4.1
HC-002	Central Square	Central	Q2 2024	01/04/2024	647	31	£5,498	25	4.3
HC-003	Central Square	Central	Q3 2024	01/07/2024	689	30	£5,857	25	4.4
HC-004	Central Square	Central	Q4 2024	01/10/2024	495	27	£4,208	25	4.0
HC-005	Victoria Park	North	Q1 2024	01/01/2024	318	35	£2,862	15	3.9

The full dataset has 42 records (rows 2 to 43 on the `Hire_Data` sheet). All the tasks in this pack use this file.

The Skills

Each chunk follows the same pattern: a short scenario, something to try, definitions of new words, and a question to check yourself at the end.

1

Opening the File

Priya has just emailed you

SCENARIO

Priya has sent you an email: "FYI — hire data attached. Please confirm row count before the 3pm meeting."

You save the attachment and open it in Excel. A yellow bar appears at the top of the window.

When you open a file downloaded from an email or a website, Excel shows a yellow bar at the top of the window. This bar is called Protected View.

WORD TO KNOW

Protected View

A safety mode that opens the file as read-only. It stops Excel from running any hidden programs in the file until you say the file is safe. Click the Enable Editing button in the yellow bar to leave Protected View and work normally.

TRY IT

Open `urban_cycle_hire_aggregated.xlsx` from the `data_files` folder. If the yellow Protected View bar appears, click Enable Editing. You can now scroll and interact with the file.

CHECK YOURSELF

What does the yellow Protected View bar ask you to do before you can work in the file?

Answer: Click Enable Editing. Until you do, you can see the file but you cannot scroll freely or change anything.

2

The Sheet Tabs

Is there more to this file?

SCENARIO

The file is open. You can see columns and rows of data. But you are not sure how the file is organised.

You wonder: are there other sheets — perhaps a set of notes, or a data dictionary?

Look at the very bottom of the Excel window. You will see small labels with names on them. These are sheet tabs. Each tab is a separate sheet inside the same file. Click a tab to switch to that sheet.

WORD TO KNOW

Sheet tab

A small label at the bottom of the Excel window. It shows the name of one sheet in the workbook. Click it to open that sheet.

WORD TO KNOW

Workbook

The full Excel file. One workbook can contain one sheet or many sheets. The GreenWheels file has two.

TRY IT

Look at the bottom of the screen. Count the sheet tabs. Write down their names.

CHECK YOURSELF

How many sheets are in the GreenWheels file, and what are they called?

Answer: Two sheets — Hire_Data and Data_Dictionary.

3

The Name Box and Ctrl+Home

Getting back to the beginning

SCENARIO

You have been scrolling around the Hire_Data sheet. You want to get back to the top-left corner quickly to check the column headers in row 1.

There are two ways to jump to cell A1.

The first way: press Ctrl and Home at the same time. Excel jumps straight to cell A1.

The second way uses the Name Box. The Name Box is a small white box at the top-left of the screen, just above column A. It shows the address of the cell you are currently in. Click it, type any cell address, and press Enter to jump there.

Next to the Name Box is a wider bar that shows the full contents of the current cell. This is called the formula bar.

WORD TO KNOW

Cell

One single box in the spreadsheet. Every cell has an address made from a column letter and a row number. A1 = column A, row 1. B5 = column B, row 5.

WORD TO KNOW

Name Box

The small white box at the top-left of the Excel window, above column A. It shows the address of the current cell. Click it, type an address, and press Enter to jump there instantly.

WORD TO KNOW

Formula bar

The wide bar next to the Name Box. It shows the full contents of the current cell — useful when a cell contains long text or a formula that does not fit inside the cell itself.

TRY IT

Press Ctrl+Home. The Name Box should show A1. Now click the Name Box, type B1, and press Enter. You should land in the Station Name header cell.

CHECK YOURSELF

What does the Name Box show when you are in the first cell of the spreadsheet?

Answer: A1.

4**Ctrl+End and Ctrl+Shift+End***Brian wants the exact number***SCENARIO**

Priya said the file has "around 40 records". Brian from Finance has already told you — without looking at the file — that he will check your row count against his pocket calculator.

You need the exact number.

Press Ctrl+End. Excel jumps to the last cell in the sheet that contains data. Look at the Name Box — it shows the address of that cell. The row number tells you where the data ends.

WORD TO KNOW**Row**

A horizontal line of cells, numbered 1, 2, 3 downwards. Row 1 usually holds the column headers. Data rows start at row 2.

WORD TO KNOW**Column header**

The label at the top of a column (usually in row 1) that names what is in that column — for example, Station Name or Total Hires.

The last row of data is not the same as the number of records. Row 1 is the header row, not a data record. So: number of records = last row number – 1.

You can also press Ctrl+Shift+End. This selects all cells from your current position down to the last data cell. It is useful for seeing the full range of data at a glance.

WORD TO KNOW**Range**

A group of cells, written as a start address and an end address — for example A1:J43. A1 is the top-left cell; J43 is the bottom-right.

WORD TO KNOW**Ribbon**

The strip of tabs and buttons across the top of the Excel window — Home, Insert, Formulas, Data, and so on. You will not need the ribbon for the navigation tasks in this pack, but you will use it in later packs for formatting and formulas.

TRY IT

Press Ctrl+Home to go to A1. Then press Ctrl+End. Write down the address shown in the Name Box.

Now press Ctrl+Home again. Then press Ctrl+Shift+End. All cells from A1 to the last data cell are now highlighted in blue.

CHECK YOURSELF

After pressing Ctrl+End, the Name Box shows J43. How many records are in the file?

Answer: 42 records. The data ends in row 43. Row 1 is the header. $43 - 1 = 42$.

5

Widening a Column

When Excel hides the numbers

SCENARIO

You look at the Period Start Date column and see ##### in every cell. The column is too narrow to show the full date. Brian cannot check your figures if the dates are invisible.

WORD TO KNOW

Column

A vertical line of cells, labelled A, B, C across the top of the sheet. Each column holds one type of information — for example, all the station names, or all the hire counts.

WORD TO KNOW

#####

Excel shows this symbol when a column is too narrow to display the value inside the cell. Widen the column to see the full date, number, or text.

To widen a column: move your mouse to the right edge of the column header letter (the grey bar at the top showing A, B, C). When the cursor changes to a double-headed arrow (↔), double-click. Excel resizes the column to fit its widest entry. This is called AutoFit.

TRY IT

Find the Period Start Date column on the Hire_Data sheet. Move your cursor to the right edge of that column's header letter. When you see the double-headed arrow, double-click. The ##### marks should be replaced by real dates.

CHECK YOURSELF

What does ##### in a cell mean, and what is the quickest way to fix it?

Answer: The column is too narrow to show the value. Double-click the right edge of the column header letter to AutoFit the width.

6**The Data Dictionary Sheet***What does that column actually mean?***SCENARIO***You notice a column called Active Bikes. Before you report anything to Priya, you want to know what it measures.**Does it count bikes out on hire right now? Or bikes assigned to the station for the whole quarter?***WORD TO KNOW****Data dictionary**

A sheet that describes every column in the dataset. It tells you the column name, the data type (text, number, date), a plain-language description of what the column measures, and what values are valid. Checking the dictionary before you analyse data is good practice.

Click the Data_Dictionary tab at the bottom of the screen. Each row in that sheet describes one column from Hire_Data. Find the row for Active Bikes and read the Description and Accepted Values columns.

TRY IT

Click the Data_Dictionary tab. Find the row for Active Bikes. Read the Description column carefully.

CHECK YOURSELF

What does the Active Bikes column measure?

Answer: The number of bikes allocated to and active at the station during the quarter. A value of 0 means the station was not operating that quarter.

7**Matching Headers to the Dictionary***Two more columns before the meeting***SCENARIO***Priya's meeting is in 15 minutes. She will ask about two more columns: Total Hires and Customer Satisfaction Score.**You want to be ready with a clear, one-sentence description of each.*

Switch back to Hire_Data. Look at the column headers in row 1. Then switch to Data_Dictionary and find the matching rows. Read the descriptions.

TRY IT

Switch to Hire_Data and note the column headers. Then switch to Data_Dictionary. Find the rows for Total Hires and Customer Satisfaction Score. Write a description of each in your own words.

CHECK YOURSELF

What is the valid range for Customer Satisfaction Score?

Answer: 1.0 (very poor) to 5.0 (excellent). Typical values are 3.5 to 4.5.

Spot the Mistake

Marcus has been helping. This is fine. Everything is fine.

Marcus from the next desk has sent you this message:

"Hi! Quick FYI — I counted the rows in that GreenWheels file for you. I pressed Ctrl+End, saw the row number was 43, and counted straight from that. So there are 43 records. You can put that in your report."

"Also: do you want a 3D pie chart of the zones? I made one. Let me know. 😊"

— Marcus

Brian is already sharpening his pencil. Use the skills from this pack to check Marcus's number.

TRY IT

Open the file. Press Ctrl+Home, then Ctrl+End. Look at the Name Box. Write down the address it shows. Work out the number of records.

What has Marcus done wrong?

ANSWER

Ctrl+End lands on J43. The last row of data is row 43. Marcus read the row number as the record count — he forgot to subtract 1 for the header row.

The last row is 43. Row 1 is the header. So the number of records is $43 - 1 = 42$, not 43.

Correct answer: 42 records (rows 2 to 43). The 3D pie chart question does not need a response right now.

Mini-Mission

A two-minute end-of-pack challenge

Priya has sent a follow-up message:

"Before 3pm: three things. How many records arrived. The first and last Hire Record IDs. What Active Bikes actually means. Keep it short. Thanks."

— Priya

You have two minutes. Use only the skills from this pack.

TRY IT

Find the three answers. Then write them as three short sentences — one per answer.

ANSWER

1. The file contains 42 records (rows 2 to 43 on the Hire_Data sheet).
2. The first Hire Record ID is HC-001 (Central Square) and the last is HC-042 (Meadow Park).
3. Active Bikes is the number of bikes allocated to and operating at a station during the quarter. A value of 0 means the station was not active.

You Can Now Answer

By the end of this pack, you should be able to answer each of these questions without looking anything up. They are written from your point of view, the way you would ask them when you need help at work.

1. How do I stop Excel from blocking me when I first open a downloaded file?
2. How do I jump straight to cell A1 from anywhere in the spreadsheet?
3. How do I find out exactly how many records a file contains?
4. What is the difference between Ctrl+End and Ctrl+Shift+End?
5. What does ##### mean in a cell, and how do I fix it?
6. How do I find the Data Dictionary sheet?
7. How do I use the Data Dictionary to find out what a column actually measures?

If You Got Stuck

Here are four common places where learners get stuck, and where to go to get unstuck.

“I cannot edit the file — I can only read it.” Go back to Chunk 1. You may need to click Enable Editing in the yellow Protected View bar.

“Ctrl+End took me to a strange cell far from the data.” Excel can remember cells that contained data before you deleted it. Save and close the file, reopen it, and try Ctrl+End again.

“I do not know which row is data and which is the header.” Go back to Chunk 3. In this dataset, row 1 is always the header. Data starts in row 2.

“I cannot see the Data_Dictionary tab.” Go back to Chunk 6. The tabs are at the very bottom of the Excel window. If the second tab is not visible, look for small arrows at the left edge of the tab bar and click the right-pointing one.